Totvs

Tangibilizing Totvs AI Agents: Learnings From Universo Totvs Event

Overweight

TOTS3.SA, TOTS3 BZ
Price (16 Oct 26):R$40.29

LatAm Technology & Telecom / Education

Marcelo Santos, CFA [AC]
(55-11) 4950-3708
marcelo.p.santos@jpmorgan.com

Livea Mizobata
(55-11) 4950-3604
livea.mizobata@jpmorgan.com

Victoria Antonello
(55-11) 4950-3703
victoria.antonello@jpmorgan.com
Banco J.P. Morgan S.A.

In this report, we draw upon our learnings from Universo Totvs industry event (see our feedback from the investor day, which took place inside the event) to provide a more tangible view of what exactly are the agents that Totvs intends to sell under the task-as-a-service (TaaS) model. We were able to distinguish 19 agents that were showcased either in the presentations or in the demonstrations stands.

- **Agents are being embedded natively across nearly every product line.** Totvs is putting AI into credit, invoicing, purchasing, collections, MRP planning, tax classification, cost closing, accounting review, contracts, payables, warehouse operations, enrollment, and payment reconciliation. LIA (AI layer of the logistics suite) stands out as a general-purpose layer. It's connected to more than 5k APIs, and clients will be able to build their own agents on it.
- **Most of the agents recommend actions and a person approves them.** For example, the Credit Release Agent hands hard cases to a credit analyst. The Supplier Suggestion Agent adds suppliers to a quote but does not pick the winner. The tax-classification and cost-closing agents propose fixes for a user to review or approve in one click. Users can often write rules in plain language. A few agents act more on their own: the Negotiation Agent collects overdue payments around the clock by email or WhatsApp, and the Accounts Payable Agent can run the payment process end to end.
- **Agents seem to be at very different stages.** The Cash Flow Projection Insight has been in production since 2023, and the Purchase Importer is also already live. Many others are pilots: the credit, invoicing, and supplier agents are expected to be generally available in 1Q27, the retail agents are planned for 2027, and several RM agents are still in development.
- **Agents vs traditional automation: autonomy, judgment, learning.** Totvs distinguishes an AI agent from normal automation by whether the logic is fixed/deterministic (a simple rule or data comparison, which is just automation) versus intelligent and adaptive — capable of reasoning over varied, complex, and changing situations to make decisions and act, rather than following one static comparison path. Moreover, agents learn and personalize over time (memory, customizable prompts/rules, confidence scoring on suggested actions).
- **Not all AI tools are agents: insights and assistants.** In addition to the agents, which act on the software and are billed task completion, we also saw two other AI-driven service types, billed as a subscription: (1) insights, offering predictive and proactive anomaly/trend detection; (2) assistants, which are reactive tools that help users find information. In this report we describe 4 insights and 6 assistants.

Table 1: LatA Tech Valuation

P/E	27E	28E
LWSA	12.6	11.1
INTB	9.3	8.9
SONDA	8.3	6.1
TOTVS	18.5	15.8
VTEX	17.0	15.0
MELI	36.6	26.8

Source: J.P. Morgan Estimates.

See page 5 for analyst certification and important disclosures, including non-US analyst disclosures.

Marcelo Santos, CFA [AC]
(55-11) 4950-3708
marcelo.p.santos@jpmorgan.com

Agents

These are systems with enough autonomy and reasoning to interpret context and independently choose and execute the best actions toward a goal, needing only guardrails (”prompts”) rather than step-by-step instructions. As per Totvs’ billing model, clients pay per completed task. We describe below what we were able to learn from the presentation tracks and product demonstrations we observed in the Universo Totvs event.

- **Supplier Suggestion Agent.** Semi-autonomous purchasing agent used during quotation. The agent suggests suppliers, including ones outside the company’s own database, from a TOTVS data lake which holds references for thousands of suppliers. Each suggestion includes a reason based on the purchasing policy and on criteria such as punctuality. It does not choose the winner; it adds participants to the quote. We found this to be one of the most interesting agents, taking advantage that Totvs software covers ~25% of the Brazilian economy.
- **Product Tax Classification Agent.** It reads the product database, compares descriptions with the government’s updated table, and suggests the NCM (Mercosur Common Nomenclature) and the C-Class Trib (tax classification code) for a person to review. Those codes determine tax benefits and the correct tax under the reform.
- **Credit Release Agent.** It analyzes sales orders blocked for credit and decides whether they can be released, using policies and limits that can be written in natural language — for example, tolerating a delay of a set number of days, or releasing a customer with a history of on-time cash payments. It runs in the background, returns an analysis and a recommendation, and calls in a credit analyst when its parameters are not enough to decide. The presenter suggested it could cut the time to approve a request by more than 70%.
- **N1 Service Agent.** Autonomous, cross-segment agent on TOTVS Atendimento, usable in back office, HR, education, and construction. It reads the service history, classifies a first-level ticket, routes the workflow, and suggests a reply.
- **Invoicing Agent.** Invoices sales orders ready for billing and can issue the tax document for quick orders. Rules can be predefined or written in natural language — for example, invoice one region first, or group orders by region — and the agent can run on a schedule.
- **Purchase Importer Agent.** It enters electronic purchase documents, links them to purchase orders, and suggests the right order when they diverge. If there is no order, one can be created and linked. It also creates the product from the supplier’s item and splits the line by cost center, warehouse, and lot.

MRP (Material Requirements Planning) is a computer-based production planning and inventory control system works out which materials and components a manufacturer needs, how many, and when, so production schedules can be met.

- **MRP Planning Agent.** After a MRP run it reads the event log, explains it, and finds production orders and purchase requests to bring forward, postpone, or cancel. It triages on its own, dropping documents already in progress, and the planner can adjust the rules in a chat — for example, deselect every purchase order. After confirmation it updates the documents in a batch.
- **Negotiation Agent.** It negotiates overdue receivables with clients around the clock by email or WhatsApp, within parameters the user sets, to recover outstanding payments and reduce the allowance for doubtful accounts.
- **Store High Performance Agent.** It reads every salesperson in a store or a group of stores and points out who cross-sells well and which categories are being ignored, so

the manager can coach and set an action plan.

- **Cross-sell Supporter Agent.** Protheus retail agent that suggests related products so the salesperson can offer them and raise the average ticket.
- **Electronic Document Auditor.** It watches for rejected electronic fiscal documents such as NF-e, finds the cause, recommends a correction, and can apply the fix and resubmit after the user approves. A dashboard shows rejection patterns so the underlying process can be fixed before the next rejection.
- **Cost Closing Agent.** Manufacturing agent for the monthly cost close. It scans the cost database for errors — wrong postings, material returns out of proportion, incorrect material requisitions, missing production-order records — checks business rules, finds the cause, and proposes a correction the user can approve in one click. It can apply the correction once authorized. Users choose which checks run and can add their own instructions; it learns from those interactions.
- **Maintenance Scheduling Agent.** It schedules technicians for the maintenance planning team: it takes maintenance orders, assigns them by specialty, and shows each technician's load.
- **Accounting Entry Agent.** It checks accounting entries for amounts and patterns that fall outside the usual range, a review that now takes a large share of the accounting team's time. It learns from feedback and suggests a cause and a fix.
- **Contract Manager Agent.** Being adapted for construction, sales, and rental contracts. It watches contracts for expirations, measurements, financial projections, price adjustments that could hurt revenue, and gaps between planned and actual measurement. It flags what is off-pattern, ranks open items by severity, and recommends a next step such as starting a renewal, drafting an amendment, ending the contract, or invoicing what is still open.
- **Accounts Payable Agent.** It runs the payables path from checking the invoice against the financial entry and the supplier, through building the payment batch (borderô), to executing the payment and processing the bank return and the reconciliation. It can stop for approval or run on its own.
- **Trend Schedule Agent.** Standard agent on the TOTVS Construção, Obras e Projetos journey. Machine learning compares the plan with what has been executed and projects how the job will move, so delays and deviations can be acted on before they land.
- **Document Validation Agent.** Standard agent for the education segment. During enrollment it checks student documents and can approve them, reject them, or send them to a person, at an accuracy level the school sets. Custom prompts can enforce a specific rule, such as requiring grades for every year of a high-school transcript.
- **Class Schedule Assembly Agent,** for the education segment. It builds a class timetable from teacher availability, course hours, and teacher history, and returns three options for someone to pick.

Marcelo Santos, CFA [AC]
(55-11) 4950-3708
marcelo.p.santos@jpmorgan.com

Other AI tools

We also learned about other AI-driven tools introduced by Totvs which do not fit the Agent/TaaS category, and are usually billed via a subscription.

Assistants

These are conversational interfaces where the user sends a message/question and receives a response, without the system autonomously executing actions on its own.

- **Fee Audit Assistant.** It compares the fees in the contract with the fees the acquirer is actually charging and flags a miss, such as a rate that was supposed to fall and did not, so the client can recover the money.
- **Data Assistant.** Assistant for questions in natural language. It handles simple requests, such as sales by client, and harder ones, such as billing anomalies, sales projections, or an answer built from several reports, by turning on specialized sub-agents.
- **Protheus Assistant.** First-level support for Protheus users, including IT. It answers questions on functions, parameters, and processes so more experienced staff are not tied up.
- **Self-Checkout Assistant.** Built into the PDV Omni self-checkout. The customer shops and places the order without a salesperson; the assistant walks them through it, and the order is ready for pickup.
- **Reconciliation Assistant.** A financial analyst can ask how reconciliation stands today, or for a given month, and get the percentages plus the items that look wrong. The presenter said the time to find a problem drops from about 50 minutes to about 5.
- **LIA.** The AI layer of the logistics suite — we are not totally sure if this is just an assistant, or also an agent, but we felt it was more assistant-like. It is connected to the data environment, the documentation, the training material, and more than five thousand APIs, so it can query any service, answer questions, build reports, and run warehouse transactions such as a separation or a conference. The platform is being extended so clients can build their own workflows and agents.

Insights

These are pre-built, structured dashboards and KPIs drawn from a dedicated analytics data layer, giving visibility into operations without requiring the user to build reports from scratch.

- **Demand Forecasting Insight.** It reads movement history, finds seasonality, and projects demand so purchasing can buy ahead.
- **Product Stockout Insight.** It uses history, seasonality, and minimum stock to project when an item will run out, so the company can replenish before the stockout.
- **Cash Flow Projection Insight.** Machine learning projects cash in and out from history, cycles, and trends, and flags a squeeze in the coming weeks or months so finance can renegotiate payments or pull receivables forward.
- **Sales Opportunity Insight.** It finds customers whose profile matches buyers of a product and suggests that product, in the style of an e-commerce recommendation, and it can open the sales order from the suggestion.

Analyst Certification: The Research Analyst(s) denoted by an "AC" on the cover of this report certifies (or, where multiple Research Analysts are primarily responsible for this report, the Research Analyst denoted by an "AC" on the cover or within the document individually certifies, with respect to each security or issuer that the Research Analyst covers in this research) that: (1) all of the views expressed in this report accurately reflect the Research Analyst's personal views about any and all of the subject securities or issuers; and (2) no part of any of the Research Analyst's compensation was, is, or will be directly or indirectly related to the specific recommendations or views expressed by the Research Analyst(s) in this report. For all Korea-based Research Analysts listed on the front cover, if applicable, they also certify, as per KOFIA requirements, that the Research Analyst's analysis was made in good faith and that the views reflect the Research Analyst's own opinion, without undue influence or intervention.

All authors named within this report are Research Analysts who produce independent research unless otherwise specified. In Europe, Sector Specialists (Sales and Trading) may be shown on this report as contacts but are not authors of the report or part of the Research Department.

Pursuant to Brazilian regulation, the primary research analyst signing this report certifies that (1) all recommendations expressed herein by the research analyst reflect the analyst's sole and exclusive personal views and have been independently produced, including from the J.P. Morgan entity in which the research analyst is an employee; and (2) the research analyst responsible for or any research analyst involved in the preparation of this report will disclose herein any situation that impacts or that could impact the impartiality of the recommendations contained in this report or that constitutes or may constitute a conflict of interest.

Important Disclosures

- **Market Maker/ Liquidity Provider:** J.P. Morgan is a market maker and/or liquidity provider in the financial instruments of/related to Totvs or related entities.
- **Client:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients: Totvs or related entities.
- **Client/Non-Investment Banking, Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-investment-banking, securities-related: Totvs or related entities.
- **Client/Non-Securities-Related:** J.P. Morgan currently has, or had within the past 12 months, the following entity(ies) as clients, and the services provided were non-securities-related: Totvs or related entities.
- **Potential Investment Banking Compensation:** J.P. Morgan expects to receive, or intends to seek, compensation for investment banking services in the next three months from Totvs or related entities.
- **Non-Investment Banking Compensation Received:** J.P. Morgan has received compensation in the past 12 months for products or services other than investment banking from Totvs or related entities.
- **Debt Position:** J.P. Morgan may hold a position in the debt securities of Totvs or related entities, if any.
- J.P. Morgan Securities LLC and/or its affiliates is acting as financial advisor to Stoneco Inc in connection with the sale of Linx and certain other software assets to TOTVS as announced on7/22/2025. The transaction is subject to regulatory approvals and shareholder vote by TOTVS. This research report and the information contained herein is not intended to provide voting advice, serve as an endorsement of the proposed transaction or result in procurement, withholding or revocation of a proxy or any other action by a security holder.

Company-Specific Disclosures: J.P. Morgan does and seeks to do business with companies covered in its research reports. As a result, investors should be aware that the firm may have a conflict of interest that could affect the objectivity of this report. Investors should consider this report as only a single factor in making their investment decision. Important disclosures, including price charts and credit opinion history tables (if applicable), are available for compendium reports and all J.P. Morgan–covered companies, and certain non-covered companies, by visiting https://www.jpmm.com/research/disclosures, calling 1-800-477-0406, or e-mailing research.disclosure.inquiries@jpmorgan.com with your request.

Marcelo Santos, CFA AC
(55-11) 4950-3708
marcelo.p.santos@jpmorgan.com

Totvs (TOTS3.SA, TOTS3 BZ) Price Chart



Date	Rating	Price (R$)	Price Target (R$)
26-Mar-24	N	30.24	36
14-Aug-24	N	29.29	37
09-Jan-25	N	26.12	34
27-Feb-25	N	35.81	38
30-May-25	N	42.80	46
22-Jul-25	NR	42.11	--
17-Dec-25	OW	45.43	59
17-Mar-26	OW	35.12	58
22-May-26	OW	32.27	57

Source: Bloomberg Finance L.P. and J.P. Morgan; price data adjusted for stock splits and dividends.
Initiated coverage Apr 05, 2006. All share prices are as of market close on the previous business day.

The chart(s) show J.P. Morgan's continuing coverage of the stocks; the current analysts may or may not have covered it over the entire period.
J.P. Morgan ratings or designations: OW = Overweight, N= Neutral, UW = Underweight, NR = Not Rated

Explanation of Equity Research Ratings, Designations and Analyst(s) Coverage Universe:
J.P. Morgan uses the following rating system: Overweight (over the duration of the price target indicated in this report, we expect this stock will outperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); Neutral (over the duration of the price target indicated in this report, we expect this stock will perform in line with the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe); and Underweight (over the duration of the price target indicated in this report, we expect this stock will underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe. NR is Not Rated. In this case, J.P. Morgan has removed the rating and, if applicable, the price target, for this stock because of either a lack of a sufficient fundamental basis or for legal, regulatory or policy reasons. The previous rating and, if applicable, the price target, no longer should be relied upon. An NR designation is not a recommendation or a rating. Some stocks under coverage have a rating but no price target; in these cases, we expect the stock will outperform/perform in line/underperform the average total return of the stocks in the Research Analyst's, or the Research Analyst's team's, coverage universe of the relevant duration of the region. In our Asia (ex-Australia and ex-India) and U.K. small- and mid-cap Equity Research, each stock's expected total return is compared to the expected total return of a benchmark country market index, not to those Research Analysts' coverage universe. If it does not appear in the Important Disclosures section of this report, the certifying Research Analyst's coverage universe can be found on J.P. Morgan's Research website, https://www.jpmorganmarkets.com.

Coverage Universe: Santos, Marcelo : Afya (AFYA), America Movil (AMX), Anima (ANIM3.SA), Cogna (COGN3.SA), Entel (ENTEL.SN), Intelbras (INTB3.SA), LWSA (LWSA3.SA), Laureate (LAUR), Megacable (MEGACPO.MX), MercadoLibre, Inc. (MELI), Millicom (TIGO), Opsimex (SITES1.MX), Ser Educacional (SEER3.SA), Sonda (SONDA.SN), TIM SA (TIMS3.SA), Telecom Argentina (TEO), Telefonica Brasil (VIVT3.SA), Televisa (TV), Totvs (TOTS3.SA), Vtex (VTEX), YDUQS (YDUQ3.SA)

J.P. Morgan Equity Research Ratings Distribution, as of October 03, 2026

	Overweight (buy)	Neutral (hold)	Underweight (sell)
J.P. Morgan Global Equity Research Coverage*	53%	35%	12%
IB clients**	83%	81%	72%
JPMS Equity Research Coverage*	51%	37%	12%
IB clients**	95%	93%	88%

***Please note that the percentages may not add to 100% because of rounding.**
****Percentage of subject companies within each of the "buy," "hold" and "sell" categories for which J.P. Morgan has provided investment banking services within the previous 12 months.**
For purposes of FINRA ratings distribution rules only, our Overweight rating falls into a buy rating category; our Neutral rating falls into a hold rating category; and our Underweight rating falls into a sell rating category. Please note that stocks with an NR designation are not included in the table above. This information is current as of the end of the most recent calendar quarter.

Equity Valuation and Risks: For valuation methodology and risks associated with covered companies or price targets for covered companies, please see the most recent company-specific research report at http://www.jpmorganmarkets.com, contact the primary analyst or your J.P. Morgan representative, or email research.disclosure.inquiries@jpmorgan.com. For material information about the proprietary models used, please see the Summary of Financials in company-specific research reports and the Company Tearsheets, which are available to download on

the company pages of our client website, http://www.jpmorganmarkets.com. This report also sets out within it the material underlying assumptions used.

History of Investment Recommendations:
A history of J.P. Morgan investment recommendations disseminated during the preceding 12 months can be accessed on the Research & Commentary page of http://www.jpmorganmarkets.com where you can also search by analyst name, sector or financial instrument.

Analysts' Compensation: The research analysts responsible for the preparation of this report receive compensation based upon various factors, including the quality and accuracy of research, client feedback, competitive factors, and overall firm revenues.

Registration of non-US Analysts: Unless otherwise noted, the non-US analysts listed on the front of this report are employees of non-US affiliates of J.P. Morgan Securities LLC, may not be registered as research analysts under FINRA rules, may not be associated persons of J.P. Morgan Securities LLC, and may not be subject to FINRA Rule 2241 or 2242 restrictions on communications with covered companies, public appearances, and trading securities held by a research analyst account.

Other Disclosures

J.P. Morgan is a marketing name for investment banking businesses of JPMorgan Chase & Co. and its subsidiaries and affiliates worldwide.

UK MIFID FICC research unbundling exemption: UK clients should refer to UK MIFID Research Unbundling exemption for details of J.P. Morgan's implementation of the FICC research exemption and guidance on relevant FICC research categorisation.

All research material made available to clients are simultaneously available on our client website, J.P. Morgan Markets, unless specifically permitted by relevant laws. Not all research content is redistributed, e-mailed or made available to third-party aggregators. For all research material available on a particular stock, please contact your sales representative.

Any long form nomenclature for references to China; Hong Kong; Taiwan; and Macau within this research material are Mainland China; Hong Kong SAR (China); Taiwan (China); and Macau SAR (China).

J.P. Morgan Research may, from time to time, write on issuers or securities targeted by economic or financial sanctions imposed or administered by the governmental authorities of the U.S., EU, UK or other relevant jurisdictions (Sanctioned Securities). Nothing in this report is intended to be read or construed as encouraging, facilitating, promoting or otherwise approving investment or dealing in such Sanctioned Securities. Clients should be aware of their own legal and compliance obligations when making investment decisions.

Digital assets, which may also be referred to as virtual or crypto assets or crypto currencies (e.g. bitcoin), discussed in this research report are subject to a rapidly changing regulatory landscape. The application of relevant laws in respect of such assets may be unsettled, and the legality of any digital asset investment may be unclear. Prior to making any investment decision regarding any digital asset, investors should seek their own legal advice and be aware of any legal considerations regarding the purchase, holding or selling of such an investment. For relevant regulatory advisories on digital assets, including bitcoin and ethereum, please see https://www.jpmorgan.com/disclosures/cryptoasset-disclosure.

The author(s) of this research report may not be licensed to carry on regulated activities in your jurisdiction and, if not licensed, do not hold themselves out as being able to do so.

Exchange-Traded Funds (ETFs): J.P. Morgan Securities LLC ("JPMS") acts as authorized participant for substantially all U.S.-listed ETFs. To the extent that any ETFs are mentioned in this report, JPMS may earn commissions and transaction-based compensation in connection with the distribution of those ETF shares and may earn fees for performing other trade-related services, such as securities lending to short sellers of the ETF shares. JPMS may also perform services for the ETFs themselves, including acting as a broker or dealer to the ETFs. In addition, affiliates of JPMS may perform services for the ETFs, including trust, custodial, administration, lending, index calculation and/or maintenance and other services.

Options and Futures related research: If the information contained herein regards options- or futures-related research, such information is available only to persons who have received the proper options or futures risk disclosure documents. Please contact your J.P. Morgan Representative or visit https://www.theocc.com/components/docs/riskstoc.pdf for a copy of the Option Clearing Corporation's Characteristics and Risks of Standardized Options or https://www.finra.org/sites/default/files/2020-08/Security_Futures_Risk_Disclosure_Statement_2020.pdf for a copy of the Security Futures Risk Disclosure Statement.

Changes to Interbank Offered Rates (IBORs) and other benchmark rates: Certain interest rate benchmarks are, or may in the future become, subject to ongoing international, national and other regulatory guidance, reform and proposals for reform. For more information, please consult: https://www.jpmorgan.com/global/disclosures/interbank_offered_rates

Private Bank Clients: Where you are receiving research as a client of the private banking businesses offered by JPMorgan Chase & Co. and its subsidiaries ("J.P. Morgan Private Bank"), research is provided to you by J.P. Morgan Private Bank and not by any other division of J.P. Morgan, including, but not limited to, the J.P. Morgan Corporate and Investment Bank and its Global Research division.

Legal entity responsible for the production and distribution of research: The legal entity identified below the name of the Reg AC Research Analyst who authored this material is the legal entity responsible for the production of this research. Where multiple Reg AC Research Analysts authored this material with different legal entities identified below their names, these legal entities are jointly responsible for the production of this research. Where more than one legal entity is listed under an analyst's name, the first legal entity is responsible for the production unless stated otherwise. Research Analysts from various J.P. Morgan affiliates may have contributed to the production of this material but may not be licensed to carry out regulated activities in your jurisdiction (and do not hold themselves out as being able to do so). Unless otherwise stated below in the legal entity disclosures, this material has been distributed by the legal entity responsible for production, or where more than one legal entity is listed under the analyst's name, the first legal entity will be responsible for distribution. If you have any queries, please contact the relevant Research Analyst in your jurisdiction or the entity in your jurisdiction that has distributed this research material.

Legal Entities Disclosures and Country-/Region-Specific Disclosures:
Argentina: JPMorgan Chase Bank N.A Sucursal Buenos Aires is regulated by Banco Central de la República Argentina ("BCRA"- Central Bank of Argentina) and Comisión Nacional de Valores ("CNV"- Argentinian Securities Commission - ALYC y AN Integral N°51).

Australia: J.P. Morgan Securities Australia Limited ("JPMSAL") (ABN 61 003 245 234/AFS Licence No: 238066) is regulated by the Australian Securities and Investments Commission and is a Market Participant of ASX Limited, a Clearing and Settlement Participant of ASX Clear Pty Limited and a Clearing Participant of ASX Clear (Futures) Pty Limited. This material is issued and distributed in Australia by or on behalf of JPMSAL only to "wholesale clients" (as defined in section 761G of the Corporations Act 2001). A list of all financial products covered can be found by visiting https://www.jpmm.com/research/disclosures. J.P. Morgan seeks to cover companies of relevance to the domestic and international investor base across all Global Industry Classification Standard (GICS) sectors, as well as across a range of market capitalisation sizes. If applicable, in the course of conducting public side due diligence on the subject company(ies), the Research Analyst team may at times perform such diligence through corporate engagements such as site visits, discussions with company representatives, management presentations, etc. Research issued by JPMSAL has been prepared in accordance with J.P. Morgan Australia's Research Independence Policy which can be found at the following link: J.P. Morgan Australia - Research Independence Policy.

Brazil: Banco J.P. Morgan S.A. is regulated by the Comissao de Valores Mobiliarios (CVM) and by the Central Bank of Brazil. Ombudsman J.P. Morgan: 0800-7700847 / 0800-7700810 (For Hearing Impaired) / ouvidoria.jp.morgan@jpmchase.com.

Canada: J.P. Morgan Securities Canada Inc. is a registered investment dealer, regulated by the Canadian Investment Regulatory Organization and the Ontario Securities Commission and is the participating member on Canadian exchanges. This material is distributed in Canada by or on behalf of J.P.Morgan Securities Canada Inc.

Chile: Inversiones J.P. Morgan Limitada is an unregulated entity incorporated in Chile.

China: J.P. Morgan Securities (China) Company Limited has been approved by CSRC to conduct the securities investment consultancy business.

Colombia: Banco J.P. Morgan Colombia S.A. is supervised by the Financiera de Colombia (SFC). Any reference in this material to products or services offered abroad by entities other than the Bank in Colombia is included exclusively for descriptive purposes. Such references do not constitute, and should not be construed as, promotional activity or the provision of financial products or services within Colombian territory, as defined under applicable Colombian regulation.

Dubai International Financial Centre (DIFC): JPMorgan Chase Bank, N.A., Dubai Branch is regulated by the Dubai Financial Services Authority (DFSA) and its registered address is Dubai International Financial Centre - The Gate, West Wing, Level 3 and 9 PO Box 506551, Dubai, UAE. This material has been distributed by JP Morgan Chase Bank, N.A., Dubai Branch to persons regarded as professional clients or market counterparties as defined under the DFSA rules.

European Economic Area (EEA): Unless specified to the contrary, research is distributed in the EEA by J.P. Morgan SE ("JPM SE"), which is authorised as a credit institution by the Federal Financial Supervisory Authority (Bundesanstalt für tungsaufsicht, BaFin) and jointly supervised by the BaFin, the German Central Bank (Deutsche Bundesbank) and the European Central Bank (ECB). JPM SE is a company headquartered in Frankfurt with registered address at TaunusTurm, Taunustor 1, Frankfurt am Main, 60310, Germany. The material has been distributed in the EEA to persons regarded as professional investors (or equivalent) pursuant to Art. 4 para. 1 no. 10 and Annex II of MiFID II and its respective implementation in their home jurisdictions ("EEA professional investors"). This material must not be acted on or relied on by persons who are not EEA professional investors. Any investment or investment activity to which this material relates is only available to EEA relevant persons and will be engaged in only with EEA relevant persons.

Hong Kong: J.P. Morgan Securities (Asia Pacific) Limited (CE number AAJ321) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission in Hong Kong, and J.P. Morgan Broking (Hong Kong) Limited (CE number AAB027) is regulated by the Securities and Futures Commission in Hong Kong. JP Morgan Chase Bank, N.A., Hong Kong Branch (CE Number AAL996) is regulated by the Hong Kong Monetary Authority and the Securities and Futures Commission, is organized under the laws of the United States with limited liability. Where the distribution of this material is a regulated activity in Hong Kong, the material is distributed in Hong Kong by or through J.P. Morgan Securities (Asia Pacific) Limited and/or J.P. Morgan Broking (Hong Kong) Limited.

India: J.P. Morgan India Private Limited (Corporate Identity Number - 68724), having its registered office at J.P. Morgan Tower, Off. C.S.T. Road, Kalina, Santacruz - East, Mumbai – 400098, is registered with the Securities and Exchange Board of India (SEBI) as a 'Research Analyst' having registration number INH000001873. J.P. Morgan India Private Limited is also registered with SEBI as a member of

the National Stock Exchange of India Limited and the Bombay Stock Exchange Limited (SEBI Registration Number – INZ000239730) and as a Merchant Banker (SEBI Registration Number - MB/INM000002970). Telephone: 91-22-6157 3000, Facsimile: 91-22-6157 3990 and Website: http://www.jpmipl.com. JPMorgan Chase Bank, N.A. - Mumbai Branch is licensed by the Reserve Bank of India (RBI) (Licence No. 53/ Licence No. BY.4/94; SEBI - IN/CUS/014/ CDSL : IN-DP-CDSL-444-2008/ IN-DP-NSDL-285-2008/ INBI00000984/ INE231311239) as a Scheduled Commercial Bank in India, which is its primary license allowing it to carry on Banking business in India and other activities, which a Bank branch in India are permitted to undertake. For non-local research material, this material is not distributed in India by J.P. Morgan India Private Limited. Compliance Officer: Ashutosh Sharma; ashutosh.j.sharma@jpmchase.com; +912261575002. Grievance Officer: Ramprasadh K, jpmipl.research.feedback@jpmorgan.com; +912261573000. Registration granted by SEBI and certification from NISM in no way guarantee performance of the intermediary or provide any assurance of returns to investors. Please visit Terms and Conditions and Most Important Terms and Conditions (MITC). The annual Compliance audit report is available at http://www.jpmipl.com/#research.

Indonesia: PT J.P. Morgan Sekuritas Indonesia is a member of the Indonesia Stock Exchange and is registered and supervised by the Otoritas Jasa Keuangan (OJK).

Korea: J.P. Morgan Securities (Far East) Limited, Seoul Branch, is a member of the Korea Exchange (KRX). JPMorgan Chase Bank, N.A., Seoul Branch, is licensed as a branch office of foreign bank (JPMorgan Chase Bank, N.A.) in Korea. Both entities are regulated by the Financial Services Commission (FSC) and the Financial Supervisory Service (FSS). For non-macro research material, the material is distributed in Korea by or through J.P. Morgan Securities (Far East) Limited, Seoul Branch.

Japan: JPMorgan Securities Japan Co., Ltd. and JPMorgan Chase Bank, N.A., Tokyo Branch are regulated by the Financial Services Agency in Japan.

Malaysia: This material is issued and distributed in Malaysia by JPMorgan Securities (Malaysia) Sdn Bhd (18146-X), which is a Participating Organization of Bursa Malaysia Berhad and holds a Capital Markets Services License issued by the Securities Commission in Malaysia.

Mexico: J.P. Morgan Casa de Bolsa, S.A. de C.V., J.P. Morgan Grupo Financiero is member of the Mexican Stock Exchange ("Bolsa Mexicana de Valores") and the Institutional Stock Exchange ("Bolsa Institucional de Valores"), and it is authorized to act as a broker dealer by the National Banking and Securities Exchange Commission ("Comisión Nacional Bancaria y de Valores").

New Zealand: This material is issued and distributed by JPMSAL in New Zealand only to "wholesale clients" (as defined in the Financial Markets Conduct Act 2013). JPMSAL is registered as a Financial Service Provider under the Financial Service providers (Registration and Dispute Resolution) Act of 2008.

Nigeria: J.P. Morgan Chase Representative Office (Nigeria) Limited is regulated by the Central Bank of Nigeria.

Philippines: J.P. Morgan Securities Philippines Inc. is a Trading Participant of the Philippine Stock Exchange and a member of the Securities Clearing Corporation of the Philippines and the Securities Investor Protection Fund. It is regulated by the Securities and Exchange Commission.

Singapore: This material is issued and distributed in Singapore by or through J.P. Morgan Securities Singapore Private Limited (JPMSS) [MDDI (P) 070/08/2026 and Co. Reg. No.: 199405335R], which is a member of the Singapore Exchange Securities Trading Limited, and/or JPMorgan Chase Bank, N.A., Singapore branch (JPMCB Singapore), both of which are regulated by the Monetary Authority of Singapore. This material is issued and distributed in Singapore only to accredited investors, expert investors and institutional investors, as defined in Section 4A of the Securities and Futures Act, Cap. 289 (SFA). This material is not intended to be issued or distributed to any retail investors or any other investors that do not fall into the classes of "accredited investors," "expert investors" or "institutional investors," as defined under Section 4A of the SFA. Recipients of this material in Singapore are to contact JPMSS or JPMCB Singapore in respect of any matters arising from, or in connection with, the material.

South Africa: J.P. Morgan Equities South Africa Proprietary Limited and JPMorgan Chase Bank, N.A., Johannesburg Branch are members of the Johannesburg Securities Exchange and are regulated by the Financial Services Conduct Authority (FSCA).

Taiwan: J.P. Morgan Securities (Taiwan) Limited is a participant of the Taiwan Stock Exchange (company-type) and regulated by the Taiwan Securities and Futures Bureau. Material relating to equity securities is issued and distributed in Taiwan by J.P. Morgan Securities (Taiwan) Limited, subject to the license scope and the applicable laws and the regulations in Taiwan. **To the extent that J.P. Morgan Securities (Taiwan) Limited produces research materials on securities not listed on the Taiwan Stock Exchange or Taipei Exchange ("Non-Taiwan Listed Securities"), these materials shall not constitute securities recommendations for the purpose of applicable Taiwan regulations, and, for the avoidance of doubt, J.P. Morgan Securities (Taiwan) Limited does not act as broker for Non-Taiwan Listed Securities.** According to Paragraph 2, Article 7-1 of Operational Regulations Governing Securities Firms Recommending Trades in Securities to Customers (as amended or supplemented) and/or other applicable laws or regulations, please note that the recipient of this material is not permitted to engage in any activities in connection with the material that may give rise to conflicts of interests, unless otherwise disclosed in the "Important Disclosures" in this material.

Thailand: This material is issued and distributed in Thailand by JPMorgan Securities (Thailand) Ltd., which is a member of the Stock Exchange of Thailand and is regulated by the Ministry of Finance and the Securities and Exchange Commission. The registered address is 548 One City Center Building, 50th Floor, Ploenchit Road, Lumphini, Pathum Wan, Bangkok 10330.

UK: Research is produced in the UK by J.P. Morgan Securities plc ("JPMS plc") which is a member of the London Stock Exchange and is authorised by the Prudential Regulation Authority and regulated by the Financial Conduct Authority and the Prudential Regulation Authority or

J.P. Morgan Markets Limited ("JPMML Ltd") which is authorised and regulated by the Financial Conduct Authority. Unless specified to the contrary, this material is distributed in the UK by JPMS plc and is directed in the UK only to: (a) persons having professional experience in matters relating to investments falling within article 19(5) of the Financial Services and Markets Act 2000 (Financial Promotion) (Order) 2005 ("the FPO"); (b) persons outlined in article 49 of the FPO (high net worth companies, unincorporated associations or partnerships, the trustees of high value trusts, etc.); or (c) any persons to whom this communication may otherwise lawfully be made; all such persons being referred to as "UK relevant persons". This material must not be acted on or relied on by persons who are not UK relevant persons. Any investment or investment activity to which this material relates is only available to UK relevant persons and will be engaged in only with UK relevant persons. A description of J.P. Morgan EMEA's policy for prevention and avoidance of conflicts of interest related to the production of Research can be found at the following link: J.P. Morgan EMEA - Research Independence Policy.

U.S.: J.P. Morgan Securities LLC ("JPMS") is a member of the NYSE, FINRA, SIPC, and the NFA. JPMorgan Chase Bank, N.A. is a member of the FDIC. Material published by non-U.S. affiliates is distributed in the U.S. by JPMS who accepts responsibility for its content.

General: Additional information is available upon request. The information in this material has been obtained from sources believed to be reliable. While all reasonable care has been taken to ensure that the facts stated in this material are accurate and that the forecasts, opinions and expectations contained herein are fair and reasonable, JPMorgan Chase & Co. or its affiliates and/or subsidiaries (collectively J.P. Morgan) make no representations or warranties whatsoever to the completeness or accuracy of the material provided, except with respect to any disclosures relative to J.P. Morgan and the Research Analyst's involvement with the issuer that is the subject of the material. Accordingly, no reliance should be placed on the accuracy, fairness or completeness of the information contained in this material. There may be certain discrepancies with data and/or limited content in this material as a result of calculations, adjustments, translations to different languages, and/or local regulatory restrictions, as applicable. These discrepancies should not impact the overall investment analysis, views and/or recommendations of the subject company(ies) that may be discussed in the material. Artificial intelligence tools may have been used in the preparation of this material, including assisting in data analysis, pattern recognition, and content drafting for research material. J.P. Morgan accepts no liability whatsoever for any loss arising from any use of this material or its contents, and neither J.P. Morgan nor any of its respective directors, officers or employees, shall be in any way responsible for the contents hereof, apart from the liabilities and that may be imposed on them by the relevant regulatory authority in the jurisdiction in question, or the regulatory regime thereunder. Opinions, forecasts or projections contained in this material represent J.P. Morgan's current opinions or judgment as of the date of the material only and are therefore subject to change without notice. Periodic updates may be provided on companies/industries based on company-specific developments or announcements, market conditions or any other publicly available information. There can be no assurance that future results or events will be consistent with any such opinions, forecasts or projections, which represent only one possible outcome. Furthermore, such opinions, forecasts or projections are subject to certain risks, uncertainties and assumptions that have not been verified, and future actual results or events could differ materially. The value of, or income from, any investments referred to in this material may fluctuate and/or be affected by changes in exchange rates. All pricing is indicative as of the close of market for the securities discussed, unless otherwise stated. Past performance is not indicative of future results. Accordingly, investors may receive back less than originally invested. This material is not intended as an offer or solicitation for the purchase or sale of any financial instrument. The opinions and recommendations herein do not take into account individual client circumstances, objectives, or needs and are not intended as recommendations of particular securities, financial instruments or strategies to particular clients. This material may include views on structured securities, options, futures and other derivatives. These are complex instruments, may involve a high degree of risk and may be appropriate investments only for sophisticated investors who are capable of understanding and assuming the risks involved. The recipients of this material must make their own independent decisions regarding any securities or financial instruments mentioned herein and should seek advice from such independent financial, legal, tax or other adviser as they deem necessary. J.P. Morgan may trade as a principal on the basis of the Research Analysts' views and research, and it may also engage in transactions for its own account or for its clients' accounts in a manner inconsistent with the views taken in this material, and J.P. Morgan is under no obligation to ensure that such other communication is brought to the attention of any recipient of this material. Others within J.P. Morgan, including Strategists, Sales staff and other Research Analysts, may take views that are inconsistent with those taken in this material. Employees of J.P. Morgan not involved in the preparation of this material may have investments in the securities (or derivatives of such securities) mentioned in this material and may trade them in ways different from those discussed in this material. This material is not an advertisement for or marketing of any issuer, its products or services, or its securities in any jurisdiction.

Confidentiality and Security Notice: This transmission may contain information that is privileged, confidential, legally privileged, and/or exempt from disclosure under applicable law. If you are not the intended recipient, you are hereby notified that any disclosure, copying, distribution, or use of the information contained herein (including any reliance thereon) is STRICTLY PROHIBITED. Although this transmission and any attachments are believed to be free of any virus or other defect that might affect any computer system into which it is received and opened, it is the responsibility of the recipient to ensure that it is virus free and no responsibility is accepted by JPMorgan Chase & Co., its subsidiaries and affiliates, as applicable, for any loss or damage arising in any way from its use. If you received this transmission in error, please immediately contact the sender and destroy the material in its entirety, whether in electronic or hard copy format. This message is subject to electronic monitoring: https://www.jpmorgan.com/disclosures/email

MSCI: Certain information herein ("Information") is reproduced by permission of MSCI Inc., its affiliates and information providers ("MSCI") ©2026. No reproduction or dissemination of the Information is permitted without an appropriate license. MSCI MAKES NO EXPRESS OR IMPLIED WARRANTIES (INCLUDING MERCHANTABILITY OR FITNESS) AS TO THE INFORMATION AND DISCLAIMS ALL LIABILITY TO THE EXTENT PERMITTED BY LAW. No Information constitutes investment advice, except for any applicable Information from MSCI ESG Research. Subject also to msci.com/disclaimer

Any mentions of third-party trademarks, brand names, products, and services are for referential purposes only and any mention thereof is not

meant to imply any sponsorship, endorsement, or affiliation.

Sustainalytics: Certain information, data, analyses and opinions contained herein are reproduced by permission of Sustainalytics and: (1) includes the proprietary information of Sustainalytics; (2) may not be copied or redistributed except as specifically authorized; (3) do not constitute investment advice nor an endorsement of any product or project; (4) are provided solely for informational purposes; and (5) are not warranted to be complete, accurate or timely. Sustainalytics is not responsible for any trading decisions, damages or other losses related to it or its use. The use of the data is subject to conditions available at https://www.sustainalytics.com/legal-disclaimers. ©2026 Sustainalytics. All Rights Reserved.

"Other Disclosures" last revised October 03, 2026.

Copyright 2026 JPMorgan Chase & Co. All rights reserved. This material or any portion hereof may not be reprinted, sold or redistributed without the written consent of J.P. Morgan. It is strictly prohibited to use or share without prior written consent from J.P. Morgan any research material received from J.P. Morgan or an authorized third-party ("J.P. Morgan Data") in any third-party artificial intelligence ("AI") systems or models when such J.P. Morgan Data is accessible by a third-party.

Completed 16 Oct 2026 06:39 PM EDT

Disseminated 19 Oct 2026 12:15 AM EDT